

of stored data, leveraging machine learning and, finally, preventive and remedial actions based on the analysis.

Pattern analytics: This offers real-time analysis and visualisation of automatically collected and correlated data to get insights into IT operations, customer experience and business outcomes. It helps to:

- Analyse rich and extensible data sets to connect the dots between IT operations, user experience and business impact
- Easily collect and correlate user, performance and business data in real-time with no code changes
- Utilise SQL on rich and extensible data sets to run *ad hoc* analysis in real-time and dig deeper into specific performance issues

These insights provide a holistic view of business process and application metrics, which gets generated from different monitoring and automation systems.

Remediation: This helps to reduce the effort put in tasks related to tickets, which can be automatically addressed via orchestrated platforms and self-service solutions. Remediation helps in:

- Reduction of execution time for tasks/processes
- Increased operational efficiency with reduced mean time to resolution

Open source AIOps platforms

Most open source AIOps projects use Python as a programming language for machine learning. Based on the enterprise requirement, various AIOps and open source tools can be combined and used on AIOps platforms. We take a brief look at the top open source AIOps platforms/tools.

SeldonIO: This open source platform deploys enterprise machine learning models on Kubernetes at a massive scale. Seldon handles scaling to thousands of production machine learning models and provides advanced machine learning capabilities out-of-the-box, including advanced metrics, request

logging, explainers, outlier detectors, A/B tests, canaries, and more.

Logpai/Loglizer: This is a machine learning-based log analysis toolkit for automated anomaly detection. Loglizer provides a toolkit that implements a number of ML-based log analysis techniques that have multiple supervised and unsupervised models.

Whylogs/Whylogs: Whylogs is an open source statistical logging library that allows data science and ML teams to effortlessly profile ML/AI pipelines and applications, producing log files that can be used for monitoring, alerts, analytics, and error analysis. It's available in Python and Java.

Jixipu/Aiopstools: This fundamental package for AIOps with Python offers the following:

- Anomaly detection
- Alarm convergence
- Time series forecasting method
- Association analysis for alarms

Log-anomaly-detector: This platform is developed using ML, and is used for log anomaly detection by connecting to streaming sources to predict abnormal log lines. It uses unsupervised machine learning models to achieve this result.

Prometheus: This open source monitoring solution simplifies pulling numerical metrics from a metrics endpoint.

Grafana: This open source metric analytics and visualisation suite is popular among Prometheus users to visualise the metrics.

Elastic Stack: This is a suite of open source products from Elastic designed to help users search, analyse, and

visualise data from any type of source, in any format, in real-time. It provides monitoring and logging solutions.

Benefits of AIOps adoption

- Improved application availability and customer satisfaction.
- Minimised application downtime on even the busiest and highest transaction days
- Expensive service disruptions are avoided, and firefighting eliminated
- Continual management of vulnerability risks. AIOps tools help to identify, analyse, prioritise and remediate vulnerability risks
- Intelligent alerts to prevent potential issues
- Increased business responsiveness
- Helps make data-driven decisions
- Helps to meet growing business demands
- Increased efficiency and optimised running costs with the help of automation and AI
- Reduced human intervention and efforts; focus on innovation

The goal of AIOps is automation, which helps in simplifying administrative tasks, thus saving time. An AIOps platform helps to scale up IT operations to support ever-growing business demands, ensuring an enhanced customer experience. It reduces the complexity of IT operations by streamlining systems, configuration management, simplifying operations and improving reliability. In short, it helps to enhance the performance of enterprise operations. **END** 

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